

Critical Thinking Assignment 4: Modifying Tables

Alexander Ricciardi

Colorado State University Global

ITS410: Database Management

Dr. Murthy Rallapalli

May 11, 2025

Critical Thinking Assignment 4: Modifying Tables

This documentation is part of the Critical Thinking 4 Assignment from ITS410: Database Management at Colorado State University Global. The documentation provides screenshots showcasing modifying tables using MySQL and the My Guitar Shop database.

The Assignment Direction:

Modifying Tables

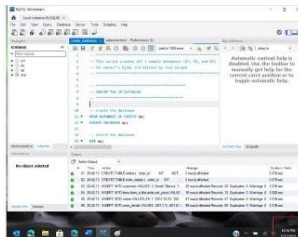
Using the My Guitar Shop database you installed in Module 1, develop the following queries.

To test whether a table has been modified correctly as you do these exercises, you can write and run an appropriate `SELECT` statement.

1. Write an `INSERT` statement that adds this row to the `Categories` table:
`category_name:` Brass
 Code the `INSERT` statement so MySQL automatically generates the `category_id` column.
 Execute the query and take a screenshot of the query and the results.
2. Write an `UPDATE` statement that modifies the row you just added to the `Categories` table. This statement should change the `category_name` column to "Woodwinds," and it should use the `category_id` column to identify the row. Execute the query and take a screenshot of the query and the results.
3. Write a `DELETE` statement that deletes the row you added to the `categories` table in exercise 1. This statement should use the `category_id` column to identify the row. Execute the query and take a screenshot of the query and the results.
4. Write an `INSERT` statement that adds this row to the `products` table:
`product_id:` The next automatically generated ID
`category_id:` 4
`product_code:` dgx_640
`product_name:` Yamaha DGX 640 88-Key Digital Piano
`description:` Long description to come
`list_price:` 799.99
`discount_percent:` 0
`date_added:` Today's date/time

Use a column list for this statement. Execute the query and take a screenshot of the query and the results

All the screenshots should show current date. Example of screenshot.



Submit your labeled results screenshots in a Word file.

Screenshots

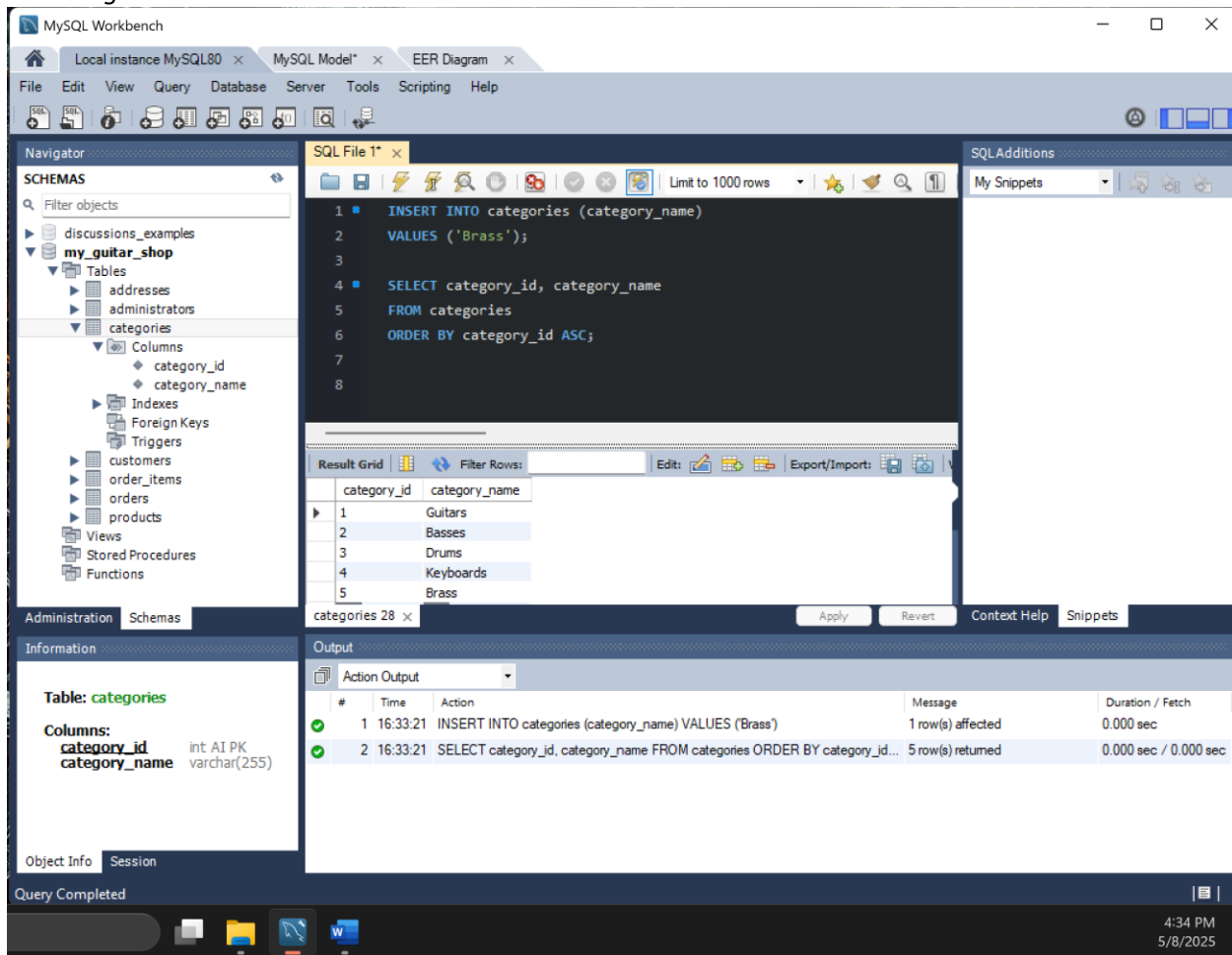
Step 1: Write an INSERT statement that adds this row to the `categories` table:

`category_name`: Brass

Note that in Figure 1 below, MySQL Workbench provides the following information about the `category_id` column: `int AI PK`

`int` stands for integer, `AI` for Auto Increment (meaning that when a new row is inserted, the new value in the column `category_id` is an auto increment by +1 of the previous `category_id` value), and `PK` stands for Primary Key.

Figure 1
Checking



Note: The figure illustrates the MySQL Workbench result after performing steps 1.

Please see the next page.

Step 2: Writing an UPDATE statement that modifies the row that was just added to the Categories table. As shown in Figure 1, that row has a category_id = 5

Figure 2
Assignment Steps 2

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'my_guitar_shop' database selected. The 'categories' table is highlighted under the 'Tables' folder. The main editor window shows the following SQL code:

```
1 UPDATE categories
2 SET category_name = 'Woodwinds'
3 WHERE category_id = 5;
4
5 SELECT category_id, category_name
6 FROM categories
7 ORDER BY category_id ASC;
```

Below the SQL editor, the 'Result Grid' shows the output of the SELECT statement:

category_id	category_name
1	Guitars
2	Basses
3	Drums
4	Keyboards
5	Woodwinds

The bottom panel shows the 'Output' tab with the 'Action Output' section. It displays the execution results of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	17:19:22	UPDATE categories SET category_name = 'Woodwinds' WHERE category_id = 5	1 row(s) affected Rows matched...	0.000 sec
2	17:19:22	SELECT category_id, category_name FROM categories ORDER BY category_id...	5 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates 'Query Completed' and the time '5:19 PM 5/8/2025'.

Note: The figure illustrates the MySQL Workbench result after performing steps 2.

Please see the next page.

Step 3: Writing a DELETE statement that deletes the row that was added to the categories table in exercise 1.

Figure 3
Assignment Step 3

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'my_guitar_shop' selected. The 'categories' table is highlighted under 'Tables'. The 'Columns' section shows 'category_id' (int AI PK) and 'category_name' (varchar(255)).

The central SQL editor contains the following queries:

```

1 DELETE FROM categories
2 WHERE category_id = 5;
3
4 SELECT category_id, category_name
5 FROM categories
6 ORDER BY category_id ASC;
7
8

```

Below the SQL editor, the 'Result Grid' shows the results of the SELECT query:

category_id	category_name
1	Guitars
2	Basses
3	Drums
4	Keyboards
NULL	NULL

The bottom panel shows the 'Output' tab with the 'Action Output' section. It displays a log of database actions:

#	Time	Action	Message	Duration / Fetch
1	17:19:22	UPDATE categories SET category_name = 'Woodwinds' WHERE category_id = 5	1 row(s) affected Rows matched...	0.000 sec
2	17:19:22	SELECT category_id, category_name FROM categories ORDER BY category_id...	5 row(s) returned	0.000 sec / 0.000 sec
3	17:20:52	DELETE FROM categories WHERE category_id = 5	1 row(s) affected	0.015 sec
4	17:20:52	SELECT category_id, category_name FROM categories ORDER BY category_id...	4 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates 'Query Completed' and the time '5:21 PM 5/8/2025'.

Note: The figure illustrates the MySQL Workbench result after performing steps 3.

Please see the next page.

Step 4: Writing an `INSERT` statement that adds this row to the `products` table. Similar to the `category_id` column in the `categories` table, the `product_id` is defined as: `int AI PK`

Figure 4
Assignment Step 4

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following SQL code:

```

1  INSERT INTO Products ( -- Using a column list
2      category_id,
3      product_code,
4      product_name,
5      description,
6      list_price,
7      discount_percent,
8      date_added
9  )
10 VALUES (
11     4, -- category_id: 4 'Keyboards'
12     'dgx_640',
13     'Yamaha DGX 640 88-Key Digital Piano',
14     'Long description to come',
15     799.99,
16     0,
17     NOW() -- Date
18 );
19
20 SELECT *
21 From Products
22 WHERE product_code = 'dgx_640';

```

The Results window shows the output of the SQL execution. The first query (INSERT) affected 1 row. The second query (SELECT) returned 1 row.

product_id	category_id	product_code	product_name	description	list_price	discount_percent	date_added
11	4	dgx_640	Yamaha DGX 640 88-Key Digital Piano	Long description to come	799.99	0.00	2025-05-08 17:51:11

The Action Output window shows the following messages:

#	Time	Action	Message	Duration / Fetch
1	17:51:11	INSERT INTO Products (-- Using a column list	1 row(s) affected	0.016 sec
2	17:55:02	SELECT * From Products WHERE product_code = 'dgx_640' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Note: The figure illustrates the MySQL Workbench result after performing steps 4.